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WHITING SCHOOL OF ENGINEERING
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Propeller-Assisted Robust 3D Hopping Robot with Hierarchical Force Allocation

Chuhan Zhang[1,2,3], Hongbo Zhang[1], Yanlin Chen[1,4], Yunxi Tang[1,4], Yun-Hui Liu[1], Mingyi Liu[2,3], and Xiangyu Chu[1,4]
Presenter: Chuhan Zhang - Johns Hopkins University, M.S.E. Robotics czhang.daniel@gmail.com ▪ chuhandanielzhang.github.io
[1]The Chinese University of Hong Kong ▪ [2]Guangdong Technion–Israel Institute of Technology ▪ [3]Technion–Israel Institute of Technology ▪ [4]Multiscale Medical Robotics Centre

SCAN: DEMO VIDEO PAPER + ARXIV
ACCEPTED PAPER
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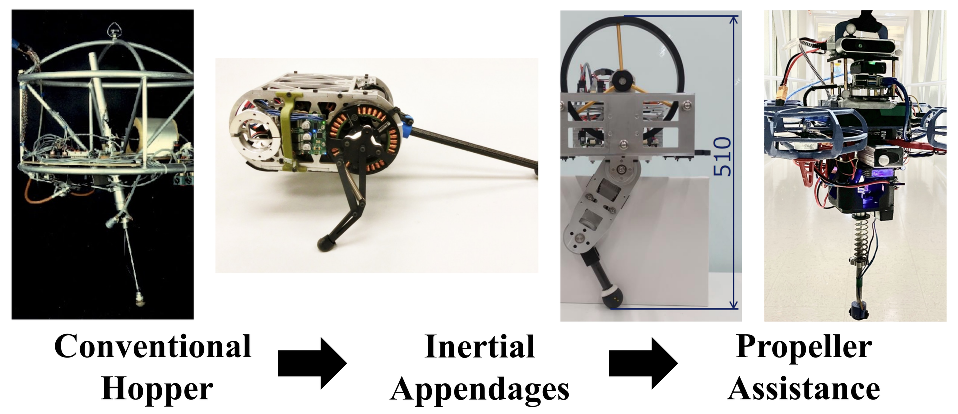
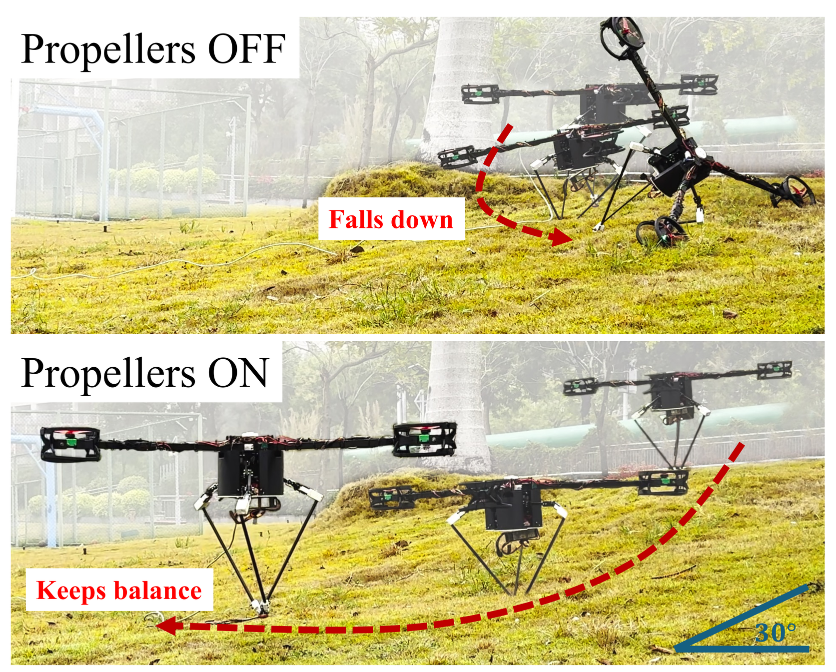
1 Problem - hybrid dynamics create an authority gap



- **Stance**
Ground reaction force exists only during a short contact interval.
- **Flight**
No contact: attitude becomes underactuated.
- **Disturbance**
Impacts and pushes inject attitude and horizontal momentum.

Control authority disappears exactly when attitude can keep drifting.
Design question: how should leg contact force and rotor thrust be coordinated across stance and flight?

2 First proof - stable on $30°$ grass

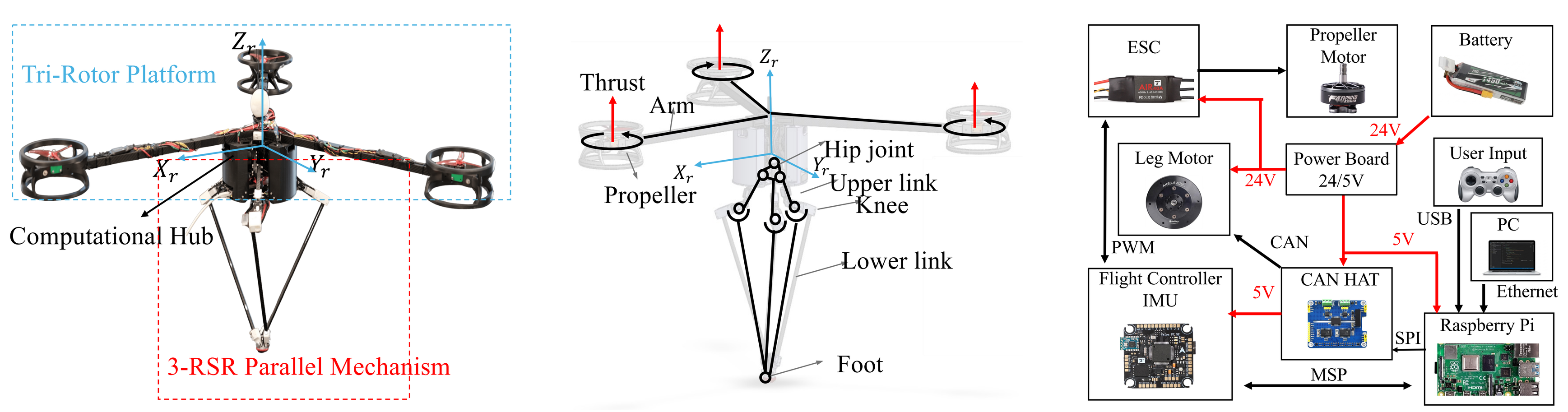


- **Propellers off**
After asymmetric contact, attitude error grows. With no flight-phase actuator, the robot falls.
- **Propellers on**
Differential thrust remains available through flight and restores attitude before the next landing.
- **What this establishes**
Residual authority in stance.
Primary attitude authority in flight.
Faster recovery from uneven impacts.

Not primary lift: the rotors close the attitude-authority gap.

Propellers off: attitude diverges and the robot falls. **Propellers on:** it remains upright and keeps hopping.

3 Platform - hardware built for leg-first hopping

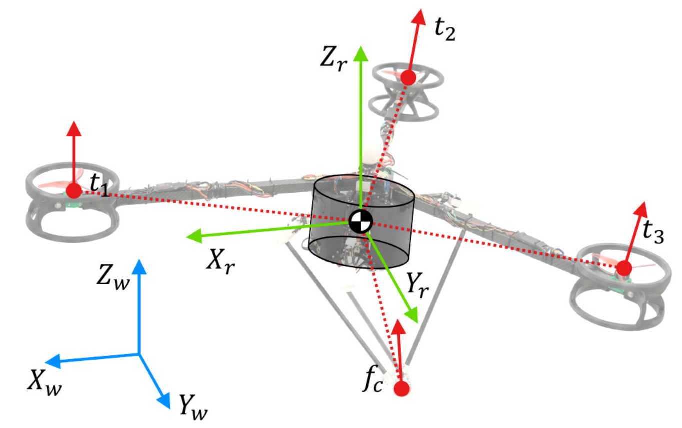


Robot design — Leg and tri-rotor components — Electronic architecture

- **3-RSR parallel leg**
Three RSR chains meet at one foot: fast 3-DoF omnidirectional placement, only **0.35 kg** moving mass.
- **External PC @ 500 Hz**
State machine, estimation, wrench planning, and the HFA allocator.
- **Computational hub**
Batteries and electronics near the hip: low center of mass, low inertia.
- **Raspberry Pi interface**
CAN to the three leg motors; MSP to the flight controller.
- **Y-layout tri-rotor**
Continuous attitude authority; **10 N** per rotor, total $0.82\,mg$: a hopper, not a drone.
- **Real-time link**
LCM over Ethernet: RTT ≈ 0.2 ms, 99% < 0.5 ms, within the 2 ms period.

3.75 kg total ▪ 9.2% leg mass ▪ 20 Nm / leg motor ▪ 500 Hz control

4 Model - one rigid body, two actuator sets



Contact force $\boldsymbol{f}_c$ and thrusts $t_{1,2,3}$ produce the body wrench.

Net wrench = leg contact + three rotors

$$\begin{bmatrix}\boldsymbol{F}_{net}\\ \boldsymbol{\tau}_{net}\end{bmatrix} = \begin{bmatrix}\boldsymbol{f}_c\\ \boldsymbol{r}_c\times\boldsymbol{f}_c\end{bmatrix} + \sum_{i=1}^{3}\begin{bmatrix}\boldsymbol{d}_t\\ \boldsymbol{r}_i\times\boldsymbol{d}_t\end{bmatrix} t_i$$

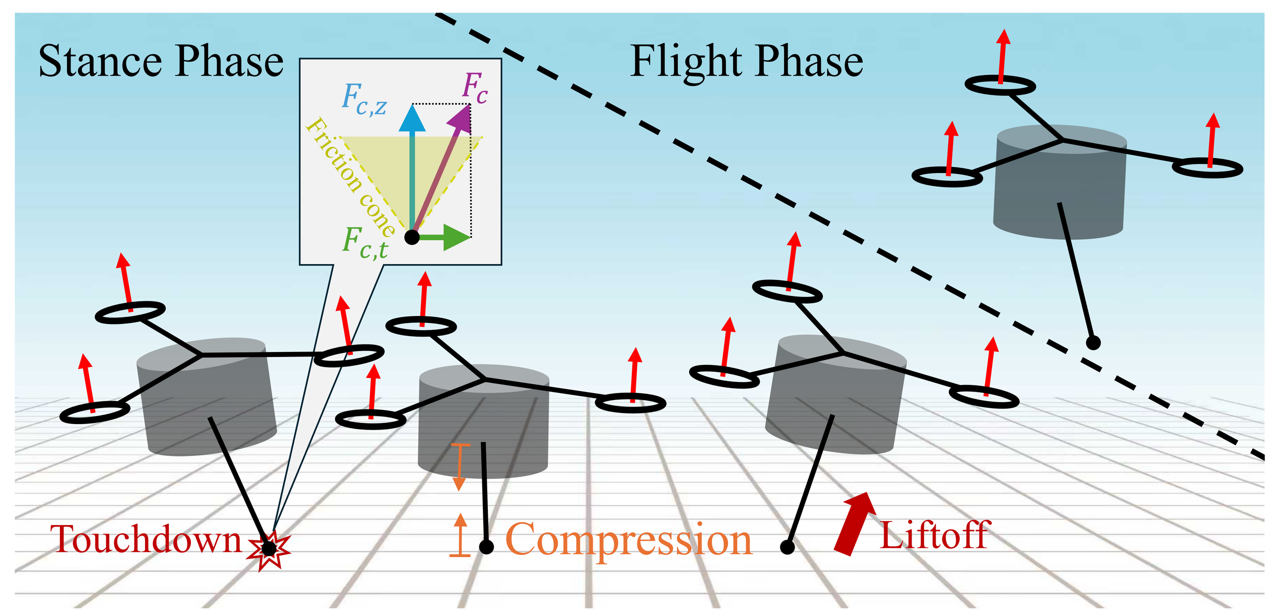
Newton-Euler dynamics of the CoM

$$m\boldsymbol{a} = \boldsymbol{F}_{net} + m\boldsymbol{g}, \quad \dot{\boldsymbol{\omega}} = \boldsymbol{I}^{-1}(\boldsymbol{R}^\top\boldsymbol{\tau}_{net} - \boldsymbol{\omega}\times\boldsymbol{I}\boldsymbol{\omega})$$

Symbols (values used in all experiments)

- $\boldsymbol{f}_c$ ground reaction force (world)
- $\boldsymbol{r}_c,\ \boldsymbol{r}_i$ arms: CoM → contact, rotor i
- $t_i \in [0, 10]$ N thrust along $\boldsymbol{d}_t = \boldsymbol{R}\boldsymbol{e}_3$
- $\boldsymbol{F}_{net},\ \boldsymbol{\tau}_{net}$ net force, moment
- m=3.75 kg; $\boldsymbol{g}$ gravity; $\boldsymbol{I}$ inertia
- $\boldsymbol{R}\in \mathrm{SO}(3)$ attitude; $\boldsymbol{\omega}$ body rate
- $\boldsymbol{J}$ leg Jacobian; τ_{max}=20 Nm limit
- μ=0.4 friction coefficient
- $p_z, \dot{p}_z$ CoM height, velocity; h_{des}=0.664 m
- l leg length; l_0=0.464 m; Δ_{td} threshold
- k_z, b_z, k_E vertical, k_v, k_r Raibert gains
- $\boldsymbol{K}_R, \boldsymbol{K}_\omega$ attitude PD gains

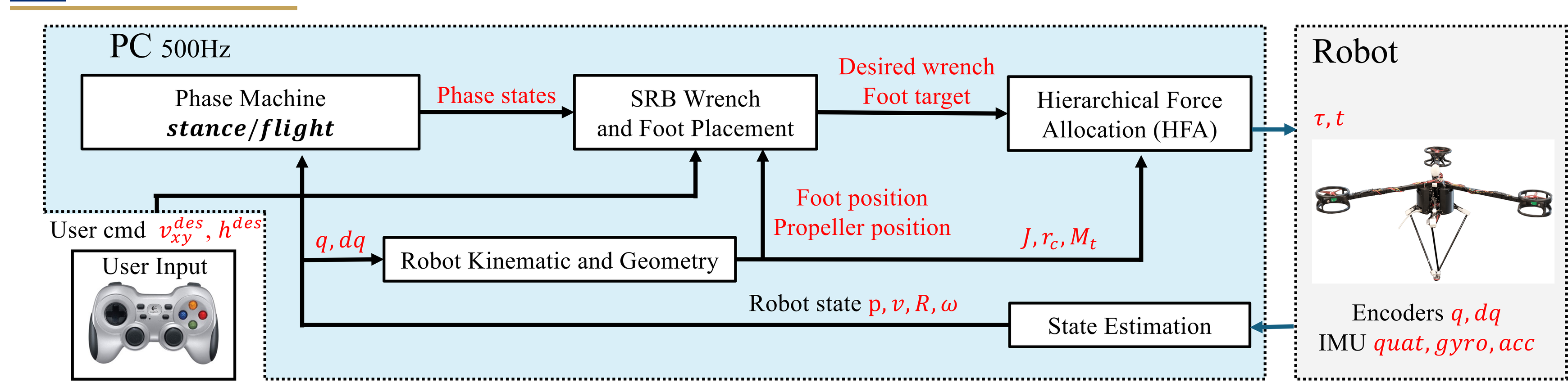
5 One hop - event-driven authority handoff



- **Touchdown:** $l \le l_0 - \Delta_{td}$
Leg absorbs impact; energy law drives the next apex.
- **Liftoff:** $l \ge l_0$
Contact wrench vanishes; rotors primary for attitude.
- **Handoff**
Events switch authority; no hardware or gain change.

Leg authority exists only in stance; rotor attitude authority is continuous through the full hop.

6 Control - leg-priority hierarchical force allocation



500 Hz architecture: commands + feedback → planner → two-stage allocator → leg torques + rotor thrusts.

A. Vertical energy hopping (stance)

$$E_{sys} = \tfrac{1}{2}m\dot{p}_z^2 + mg\,p_z, \quad E_{des} = mg\,h_{des}$$
$$f_{c,z}^{des} = \max\big(0,\ k_z(h_{des}-p_z) - b_z\dot{p}_z + k_E(E_{des}-E_{sys})\big)$$

Injects push-off energy, absorbs impact.

B. SO(3) attitude PD (both phases)

$$\boldsymbol{e}_R = \tfrac{1}{2}(\boldsymbol{R}_{des}^\top\boldsymbol{R} - \boldsymbol{R}^\top\boldsymbol{R}_{des})^\vee$$
$$\boldsymbol{\tau}_b^{des} = -\boldsymbol{K}_R\boldsymbol{e}_R + \boldsymbol{K}_\omega(\boldsymbol{\omega}^{des} - \boldsymbol{\omega})$$

Roll/pitch only; yaw left free. $\boldsymbol{\tau}^{des} = \boldsymbol{R}\boldsymbol{\tau}_b^{des}$.

C. Leg first - constrained least squares

$$\boldsymbol{f}_{c,xy}^{cmd} = \arg\min_{\boldsymbol{f}_{xy}} \left\|(\boldsymbol{r}_c\times\boldsymbol{f}_c)_{rp} - (\boldsymbol{\tau}^{des})_{rp}\right\|_2^2$$
$$\text{s.t. } \|\boldsymbol{f}_{xy}\|_2 \le \mu f_{c,z}^{des}, \quad \|\boldsymbol{J}^\top(-\boldsymbol{R}^\top\boldsymbol{f}_c)\|_\infty \le \tau_{max}$$

The leg realizes all it can under friction and torque limits.

D. Rotors take only the residual

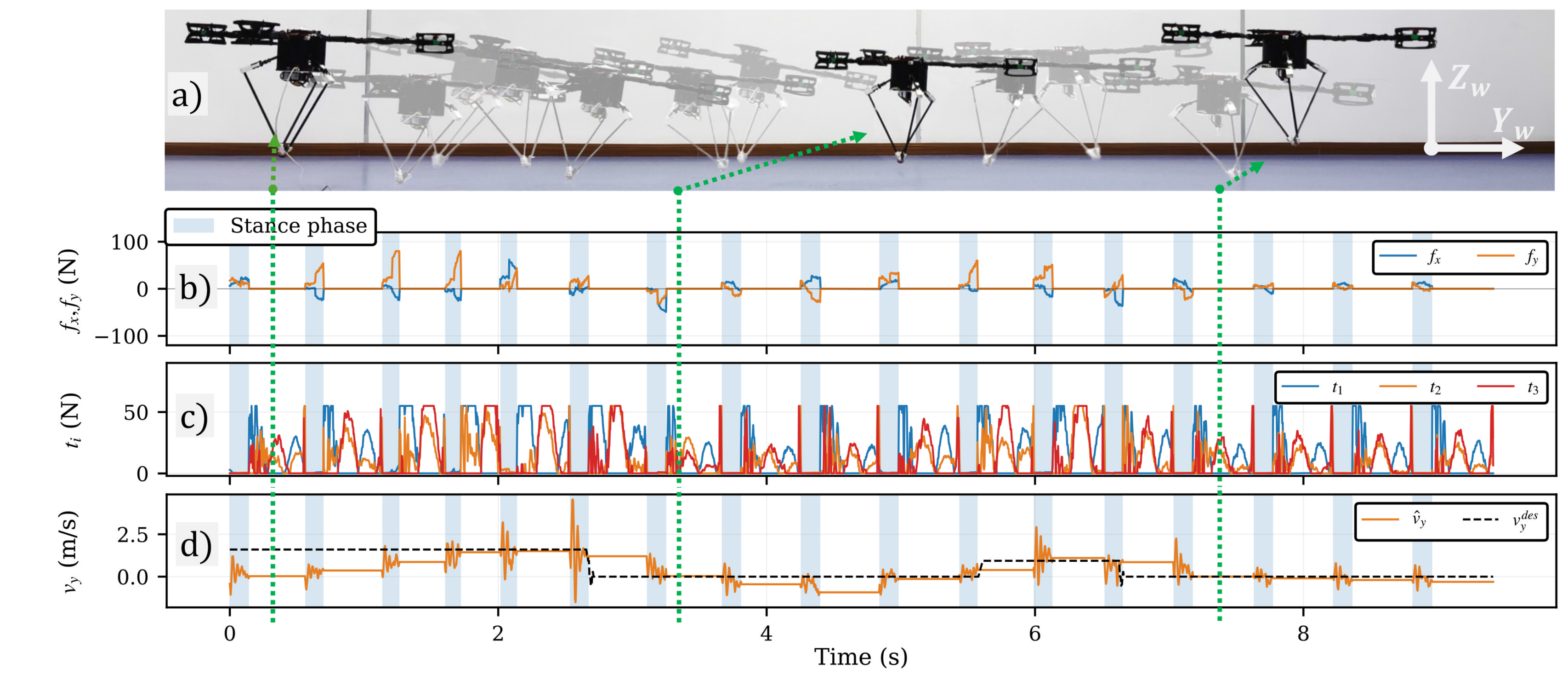
$$\boldsymbol{\tau}_{res}^{cmd} = \boldsymbol{\tau}^{des} - \boldsymbol{r}_c\times\boldsymbol{f}_c^{cmd}$$
$$\delta\boldsymbol{t}^{cmd} = \arg\min_{\delta\boldsymbol{t}} \left\|(\boldsymbol{M}_t\,\delta\boldsymbol{t})_{rp} - (\boldsymbol{\tau}_{res}^{cmd})_{rp}\right\|_2^2$$

s.t. thrust bounds; $\boldsymbol{t}^{cmd} = \frac{T_{base}}{3}\mathbf{1}_3 + \delta\boldsymbol{t}^{cmd}$, $T_{base} = 0.02\,mg$.

E. Flight - rotors primary; Raibert-style foot placement

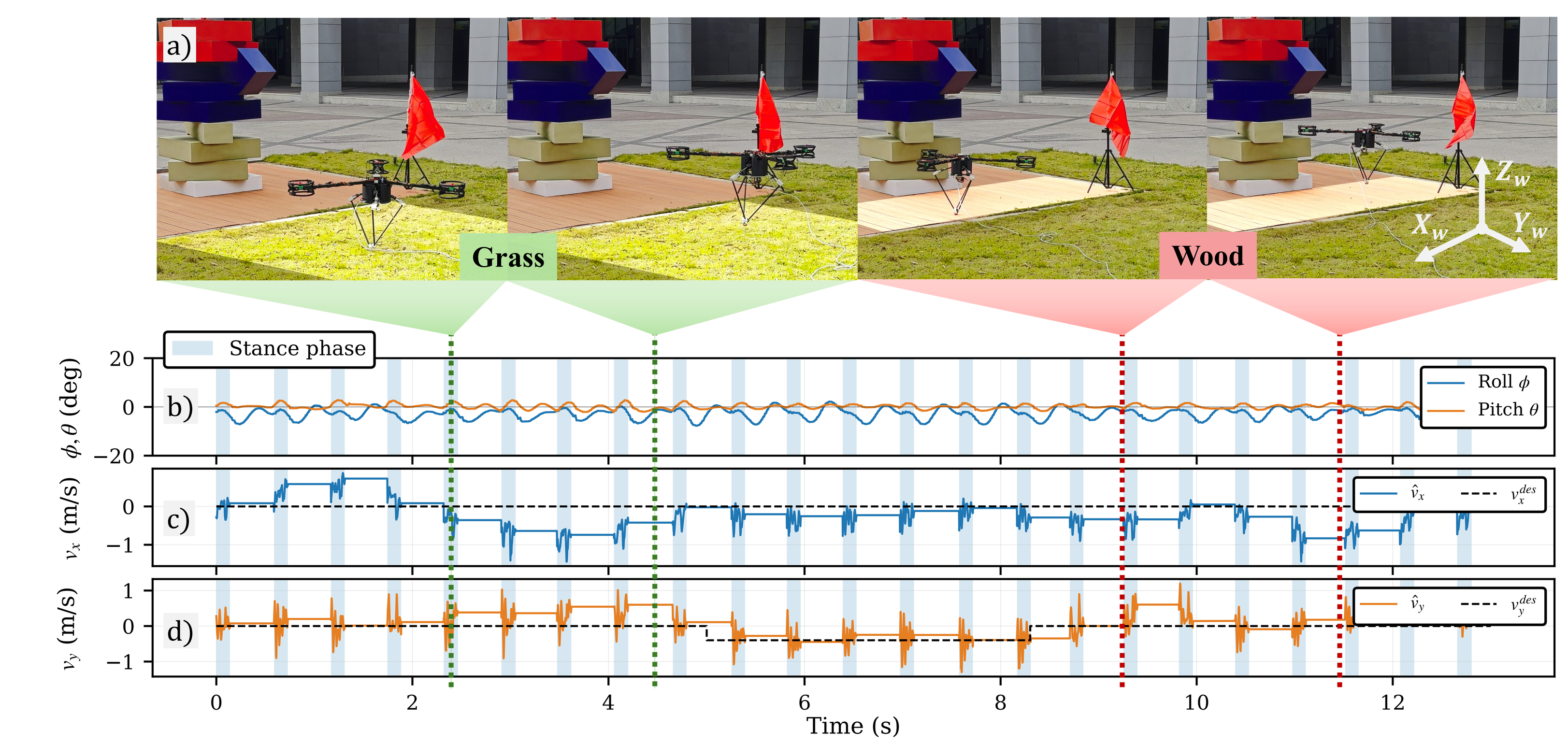
$$\boldsymbol{f}_c = \mathbf{0}; \quad \text{rotors track } (\boldsymbol{\tau}^{des})_{rp}; \quad \boldsymbol{r}_{td,xy}^{des} = k_v\boldsymbol{v}_{xy} - k_r\boldsymbol{v}_{xy}^{des}$$
$$\boldsymbol{p}_{foot}^{des} = \left[x_{td}^{des},\ y_{td}^{des},\ -\sqrt{l_0^2 - \|\boldsymbol{r}_{td,xy}^{des}\|_2^2}\right]^\top$$

7 Validation - indoor stop-and-go tracking



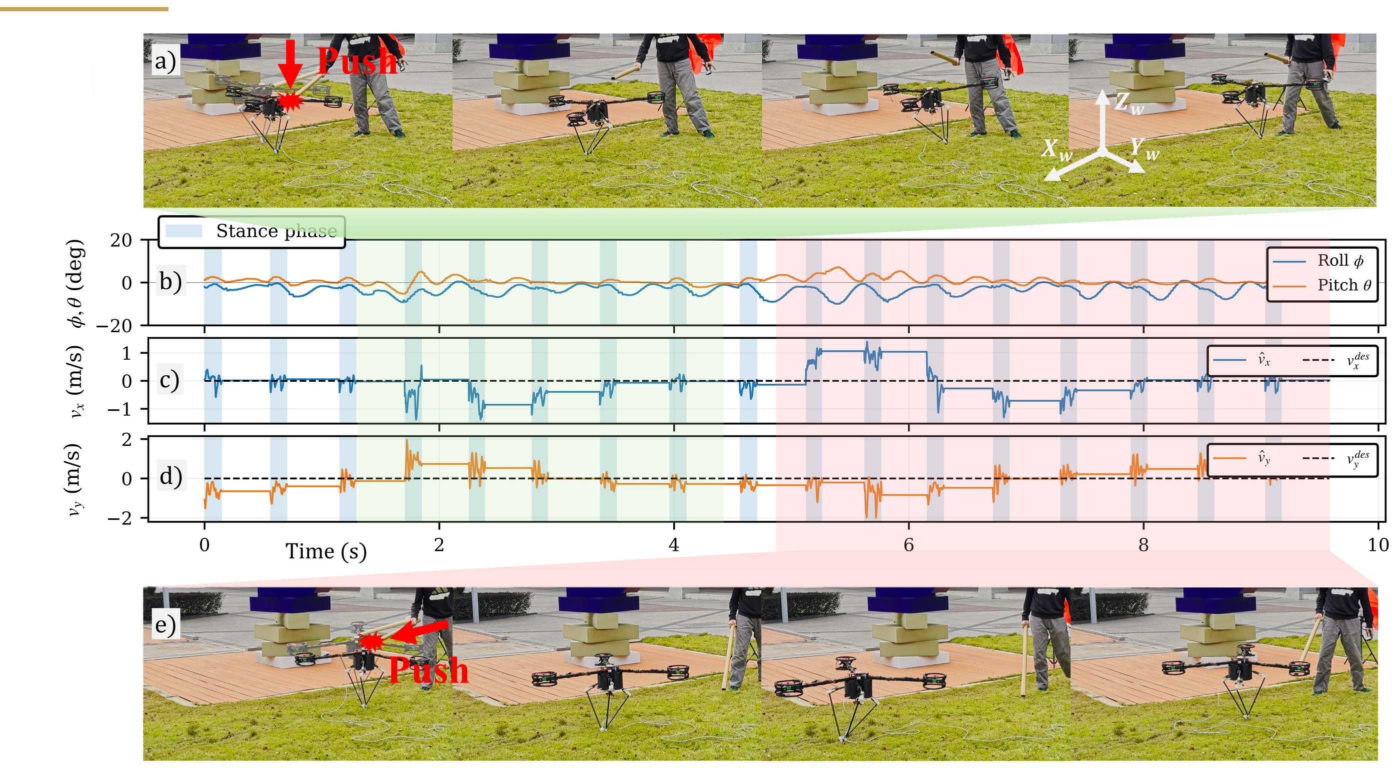
Stop-and-go commands: accelerate, then hard stop (**peaks 1.6 and 0.94 m/s**) - 17 hops.
Velocity recovered stride to stride by **Raibert-style foot placement** (Panel 6, E).

8 Validation - grass to wood



Grass → wood under light wind: **23 hops** - same controller, no terrain model, no gain retuning.
Attitude stays tight: roll $\sigma = 2.1°$, pitch $\sigma = 1.0°$; deviations decay within 1–2 hops.

9 Validation - push recovery



Rotor-arm push: propellers restore roll/pitch in flight - stable again in **1–2 hops**.
Body push (≈1.4 m/s velocity spike): foot placement recovers tracking in **3 hops**.